Course Introduction

Sehyun Kwon

Mathematical Data Science
Hanyang University ERICA

Sehyun Kwon

Experience

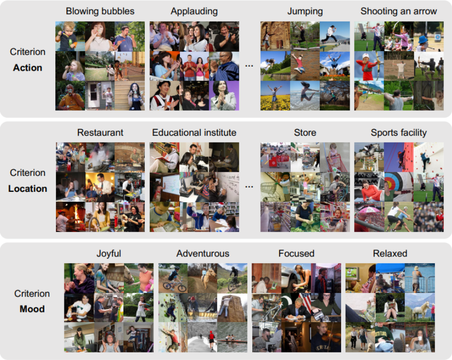
- **Assistant Professor** Fall 2026–present
  Department of Mathematical Data Science,
  Hanyang University ERICA
- **Staff Engineer** 2025–2026
  Samsung Research

Education

- **Ph.D. in Artificial Intelligence** 2025
  Seoul National University
- **M.S. in Mathematical Sciences** 2022
  Seoul National University
- **B.S. in Mathematics Education** 2020
  Dankook University

Research interests

Deep learning · Large language models · Multimodal AI



Interested in artificial intelligence?

If you want to

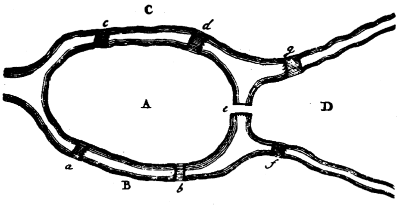
- study artificial intelligence more deeply,
- build practical skills and grow quickly as an AI engineer,
- or prepare for a career in AI,

please feel free to send me an email at any time.

I am currently recruiting undergraduate research interns.

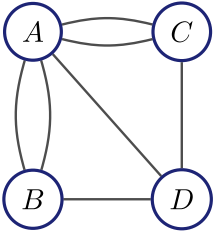
You are also welcome to contact me for an informal conversation about research, study plans, or career paths.

Discrete mathematics



A concrete problem

$\Longrightarrow$

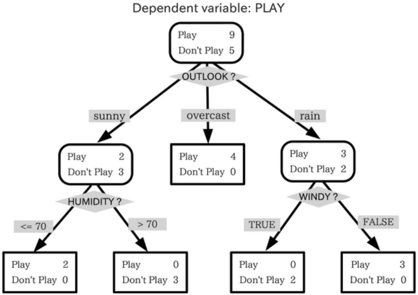


A finite structure

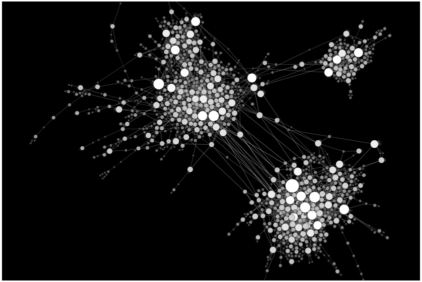
Discrete mathematics studies distinct objects and their relationships. Typical objects include integers, logical statements, arrangements, and networks.

Euler's Königsberg figure via Wikimedia Commons, public domain.

Discrete mathematics in data science



Decisions and algorithms

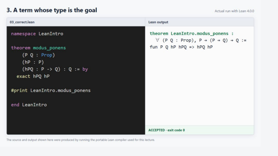


Interactions in data

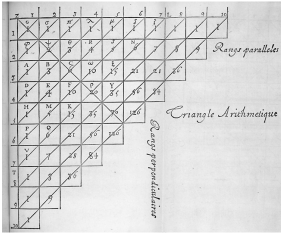
Data science and computer science repeatedly work with finite choices, symbolic rules, and networks. Discrete mathematics gives us a precise language for representing and analyzing these structures.

Decision tree: T-kita, public domain. Network: Martin Grandjean, CC BY-SA 4.0; converted to grayscale.

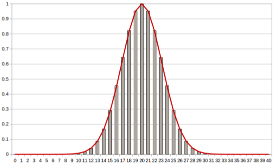
Proof, counting, and probability



Mathematical proof



Counting

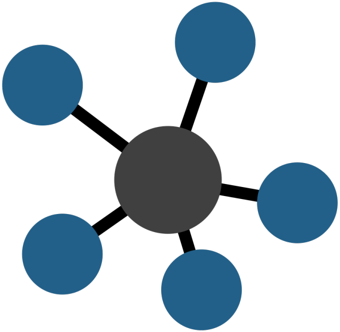


Discrete probability

We will learn how to write correct proofs, count possible configurations, and model uncertainty. These tools help us justify algorithms and reason about finite outcomes.

Pascal manuscript: Cambridge University Library via Wikimedia Commons, public domain.

Graphs and information



Graph theory

Basic information theory

Graph theory studies how objects are connected. Information theory studies how data can be encoded, measured, and transmitted reliably.

Reference materials

We will use the following books as reference materials throughout the course.

- **Kenneth H. Rosen**
  Discrete Mathematics and Its Applications, 8th edition, McGraw Hill.
- **Eric Lehman, F. Thomson Leighton, and Albert R. Meyer**
  Mathematics for Computer Science, MIT OpenCourseWare.

These references provide additional explanations, examples, and exercises for topics covered in the lectures.

Assessment

Component	Weight
Attendance	10%
Assignments	20%
Midterm replacement assignment	20%
Final exam	30%
IC-PBL Team Project	20%
Total	100%

Assignments

3-5 assignments will be given during the semester.
Each assignment is designed to help you practice ideas and methods introduced in class.
Assignments will include handwritten work and coding problems.

Midterm assignment: AI and an open problem

Choose an open problem related to discrete mathematics or data science and use AI to investigate a proof or a counterexample.
One possible starting point is OEIS A113010:

`https://oeis.org/A113010`

You may choose another open problem.
Your report should explain

- what you tried and how you used AI,
- which claims or outputs you checked critically,
- how far you progressed and where you became stuck,
- what you changed after unsuccessful attempts.

Solving the problem is *not* required. Assessment will focus on the quality of your reasoning, verification, documented trial and error, and reflection on the process.

Final exam

The final exam will be based on the lecture slides and assignments.
The exam will be **open book**.
You may bring any printed or handwritten materials on paper.
Electronic devices, including tablet PCs, are not permitted during the exam.

IC-PBL Team Project

Use discrete mathematics to address a real-world problem.
The project will be evaluated by how clearly you explain

- why the problem is important,
- why your approach is appropriate,
- how discrete mathematics is used,
- and how you analyze or solve the problem.

You will submit a written report and give a presentation.

Project topic: PageRank

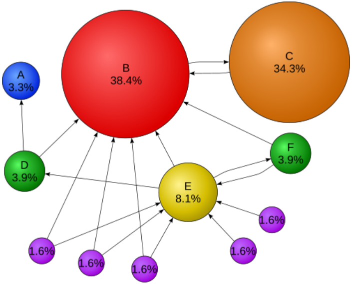
PageRank assigns a score to each object in a network. The score represents its importance based on the pattern of links in the network.

A proof of the PageRank algorithm is **not required**.

For this project, you will

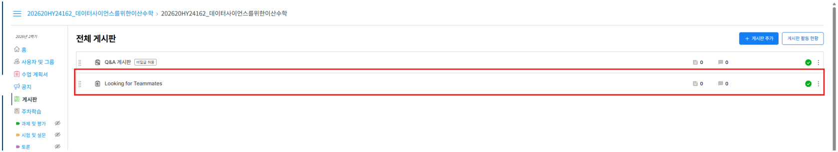
- understand what a PageRank score means,
- explore its applications,
- and use PageRank to analyze a problem of your choice.

One well-known application is ranking web pages, but you may work with any suitable network.



Finding teammates

Teams of 5–6 students will be formed near the end of September. You may find teammates through our **KakaoTalk open chat** or the **Looking for Teammates** board on the LMS.
Students who do not form a team will be assigned to one by the instructor.



AI use policy

- You may use any AI tool, in any way and to any extent, for assignments and the IC-PBL Team Project.

  Exception: AI tools are not permitted during the final exam.

- This course will provide guidelines for using AI effectively to support your learning.

- If you rely on AI without critically checking its output or understanding what you submit, you will not receive a good score for that work.